

I2R A23 CNC INDUCTION MANUAL

1. SAFETY

2. GETTING STARTED

INDUCTION MANUAL: SAFETY

The I2R A23 CNC has a 3HP (2.2kW) spindle that is operated by program control, and can rotate up to 24,000 RPM.

It has 3 powerful stepper motors that can make rapid movement of the tool in the 1-3 directions of the axes, singly, or concurrently across the whole of the CNC bed.

These factors make this machine very dangerous if not used with great care, careful planning, and adherence to the following safety instructions.

DO NOT use the I2R A23 CNC unless you:

- **Have completed induction training in the use of the I2R A23 CNC and UCNC Controller interface;**
- **Have been approved to do so, and**
- **Are working within your level of competency.**

Plan Each Job

- Carefully make your choice of stock material, especially if you are using wood, the types / species of wood that will be most suitable, and the types of cuts you need to make to that material.
- Determine if there is a need for waste material – are you intending to cut through to the bottom of the stock material.
- Determine the size and length of job – you are going to have to stay near the CNC for the whole job.
- Determine the type and size of tool(s) (router bits) needed.
- Determine the type of the clamps (stays) needed, and position of these clamps relative to the tool path.
- Determine how you will dispose of the waste material created by the tool.
- Determine what will be a Safe Height for each job, and program all non-cutting movements and pauses of the tool head at that Safe Height to avoid unintentional cuts and the tool left rotating against the stock material. The Safe Height is the height above the stock material at which the tool head can be safely moved, without any danger of cutting into the stock. For

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example, 5mm above a level surface of the stock material is generally considered to be a Safe Height.

- Read the UCNC manual so that you are familiar with its operations and know how to effectively perform each operation required for the job.
- Experiment with small scraps of the chosen material and different parameters / settings to ensure that the job is feasible, safe and the most effective way you know to get the desired results, and that the GCode is correct for the CNC and stock material required for the job.
- Consider how the tool will cut against the grain of the stock material to get the best results.
- If in doubt, discuss each job with other experienced CNC users.

Personal Safety

- Tie back long hair – DO NOT allow your hair to get near the CNC bed or the rotating tool head.
- DON'T allow any foreign material, such as cloths or cords near the CNC bed or rotating tool head
- DON'T put your hands near the rotating tool head when it is in operation. It may move suddenly and catch your hands. The rotating tool head is a very sharp, strong instrument that can cause severe injuries.
- Have someone else nearby to call if you might get into trouble, preferably someone who can provide some supervision and is familiar with the use of CNC. Don't rely on having a mobile phone in case of an emergency.
- Always be aware that the CNC tool head and gantry may move unexpectedly and have powerful motors that can trap you limbs if you are too close.
- Always operate the CNC with the exhaust fan on and the container door open to avoid a build-up and inhaling of dust and finely tooled material.

Clothing

- DON'T wear loose clothing which may get caught in the rotating tool head.
- DON'T wear dangling jewellery that may catch in the path of the tool head.
- Wear safety goggles to prevent any waste material hitting your eyes
- Wear ear protection to avoid the sustained noise of the CNC and the extractor fan
- Wear a face mask or respirator to avoid inhaling dust created by the CNC. You must use an approved respirator if you are machining any Category 2 Carcinogen wood which has a high lignum content, such as blackwood, teak, mahogany, birch and walnut. The dust from these materials are potential carcinogens, leading to nasal, sinus and lung cancers. Inhaling wood dust can also cause asthma. Check the advice with other materials, including glues, to ensure that they do not have cancerous or other toxic properties when machined.

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- Wear a full face mask and protective gloves when machining metal. The gloves are needed to protect you from the metal shavings generated by the tool, and from the heat generated by the tool rotating against the stock material.
- Wear secure footwear, not sandals, open shoes or bare feet.

Monitoring

- Visually check the progress of the job:
 - using the tool path shown on the UCCNC screen,
 - the Work Coordinates shown on the UCCNC screen, particularly the Z coordinate, and
 - watching the actual movement of the tool head.
- Listen to the sounds of the CNC. If the sounds change expectedly, then visually check the state of the CNC and screen to check if anything has gone wrong or is behaving unexpectedly.
- Don't leave the CNC job for more than a very short break – problems, including fires, can arise very quickly.

Securing the Material

- Carefully plan how you secure the stock material while it is being machined. There can be a lot of force exerted by the moving tool.
- Choose a suitable incremental step, such as 0.2 mm, for successive passes of the tool across the material, so that you do not exert too much force, and get a much finer, cleaner cut of the stock material.

Trying to cut too great a depth of material in a single face places a lot of force on the stock material, and if it moves or breaks free of the clamps it can be thrown out of the CNC.
- Ensure that the tool path avoids all clamps and their securing bolts.

Check via simulation, or via dummy run of the job at a safe height above the stock and clamps to check the extent of the tool movement and its clearance of any clamps.
- Plan the sequence of cuts to ensure that any outside cut which cuts through the whole stock material is last, and that the cut piece is firmly stuck to the underlying waste material, by screws, strong glue or strong tape.
- If uncertain about the stability of the final cut on the stock material, use **TABS**, or small pieces of material that are left around the cut object so that it stays in place until the final cut. These TABS can be manually removed later.
- Beware that when an object is fully cut out of the stock material, the object can become a projectile if not adequately secured.

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Cleaning

- Clean down the CNC before starting any job. Wipe the surfaces and the threads with a soft, lightly oiled cloth to remove any dust or other waste,
- Use the exhaust fan and hose to remove dust from the bed and in the recesses of the CNC.
- While the job is operating, use the dust shoe or the hose to constantly remove dust / waste as it is generated by the tool head.
- After a job is completed, vacuum all waste material from the CNC and wipe the surfaces and threads with a soft, lightly oiled cloth to remove any dust.
- Vacuum or thoroughly sweep the floor to remove any waste material, especially metal waste which can get into shoes, clothes and bare skin.
- Dispose of all waste in the rubbish bin, not on the ground outside the container.

Changes

- DON'T make ad hoc changes during the progress of the job, such as moving one or more of the clamps. This will result in movement of the stock and an imprecise result.
- DON'T pause the job unnecessarily. A pause and continue break will likely leave an imperfection in the finished product.

Fire

- Monitor the progress of the CNC to detect any likelihood of fire. If the tool head is causing smoke then it is running too fast or is not coping well with cutting the material. **Stop the CNC immediately by hitting the EMERGENCY STOP button on the CNC Controller.**
- If a malfunction occurs, due to a fault in the GCode, fault in the UCCNC controller, error detection by the CNC controller, or hardware failure then the tool bit may be left rotating at high speed in the same place, and can quickly heat up the stock material and burst into flames.

If a fire occurs then immediately press the EMERGENCY STOP button on the CNC Controller, and use a fire extinguisher to stop the fire. If you can't control it then call 000.

- DON'T leave the CNC running with the tool rotating in contact with stock material while you perform operations on the UCCNC screen or do any operations on the PC. If you need to check or make changes on the PC, then raise the tool head to a Safe Height before, so that the tool is not in contact with the stock material.

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- Plastics should be machined on low rotating speeds, as they can easily melt and burst into flames, or emit toxic fumes.
- Be aware of the location of the nearest fire extinguisher and its use.

Cutting Metal

- Choose tools that are hardened, and can effectively cut the metal.
- Plan how to get the best results through the type of cuts you use.
- Use a mister to dissipate the build-up of heat at the tool face.

Emergency Stops

- Be aware of where the **EMERGENCY STOP** button is located on the CNC controller, and hit it immediately in the case of a failure in the cut, or any other emergency.
- Always use the **EMERGENCY STOP** button on the CNC Controller in preference to the **RESET** button on the UCCNC screen. The **RESET** button may take longer to react and stop.
- Keep other material and tools clear of the CNC so that you have clear access to all parts of the CNC bed, and to the exhaust fan controller.
- DO NOT allow anyone else who is not authorised to use the CNC or get near to the CNC while it is in operation, or distract you from monitoring the progress of the job.
- Remember that your safety and that of anyone else in the tools container is paramount. Any job can be repeated after an Emergency Stop, and tool and equipment can be replaced.
- In the case of a serious injury call 000 and seek help immediately

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1. Start the CNC:

- a. Check the power to the CNC is On.
- b. The CNC Controller Box is located below the front right of the CNC.

Turn the green Power Switch to On (I).

Clear the **Emergency Stop** button by rotating it clockwise.



- c. Turn on the Dust Extractor Fan:
 - i. Open the switch cover beneath the Emergency Stop red switch.
 - ii. Press the green On (I) switch. The Extractor Fan should start.



2. Start the PC:

- a. If the screen is not on, press the Power On button at the bottom right of the screen.
- b. If the PC is not running, press the Power On button on the top right of the PC.
- c. When the login screen appears, enter the PIN **7008**.
- d. Open the UCCNC application on the top left of the screen.

The UCNC Controller software should automatically connect to the CNC.



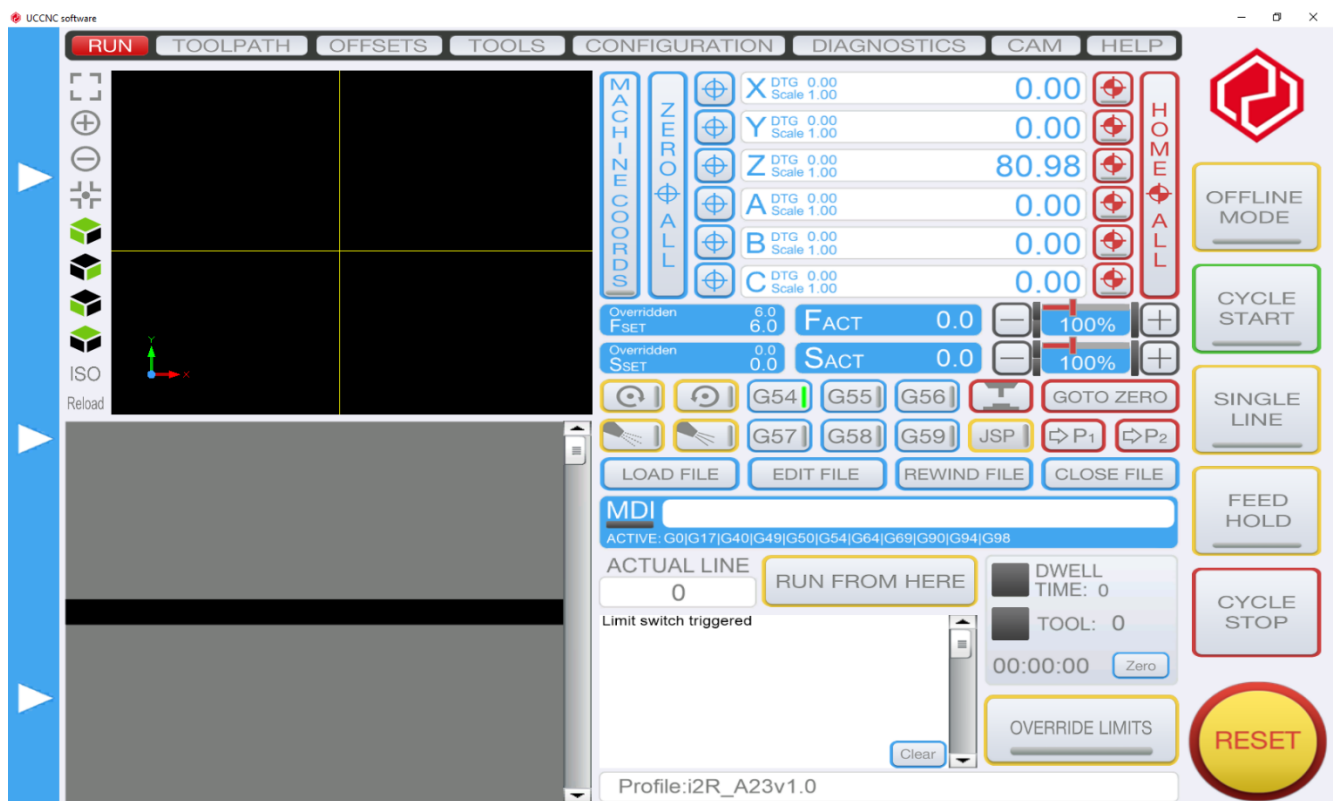
Click **OK** on the dialogue box: **WARNING – Please home the machine!!**

- e. If you are using a GCode file from a USB stick, insert the USB stick into the port at the top centre of the PC.

3. Initiate the UCCNC Controller:

- a. The **RESET** button will be flashing red and yellow.
Click the **RESET** button.

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4. The I2R A23 CNC has 2 sets of coordinates:
 - a. The Machine Coordinates (X,Y,Z) that specify the location of the tool head relative to the Machine Origin, which is located at the front left of the CNC bed.
 - b. The Work Coordinates (X,Y,Z) that specify the location of the tool head relative to the position of the head at the commencement of a job.

You must set the Machine Coordinates once when you start the CNC, by first Homing the tool head to the Machine Origin at the front left of the CNC bed.

You must set the Work Coordinates prior to the commencement of each job by positioning the tool head at the commencement position, and then zeroing one or more of the axes.

5. Set the Machine Coordinates.

The CNC operates on 3 right-handed axes:

X extends from the left side (zero) to the right side (X limit)

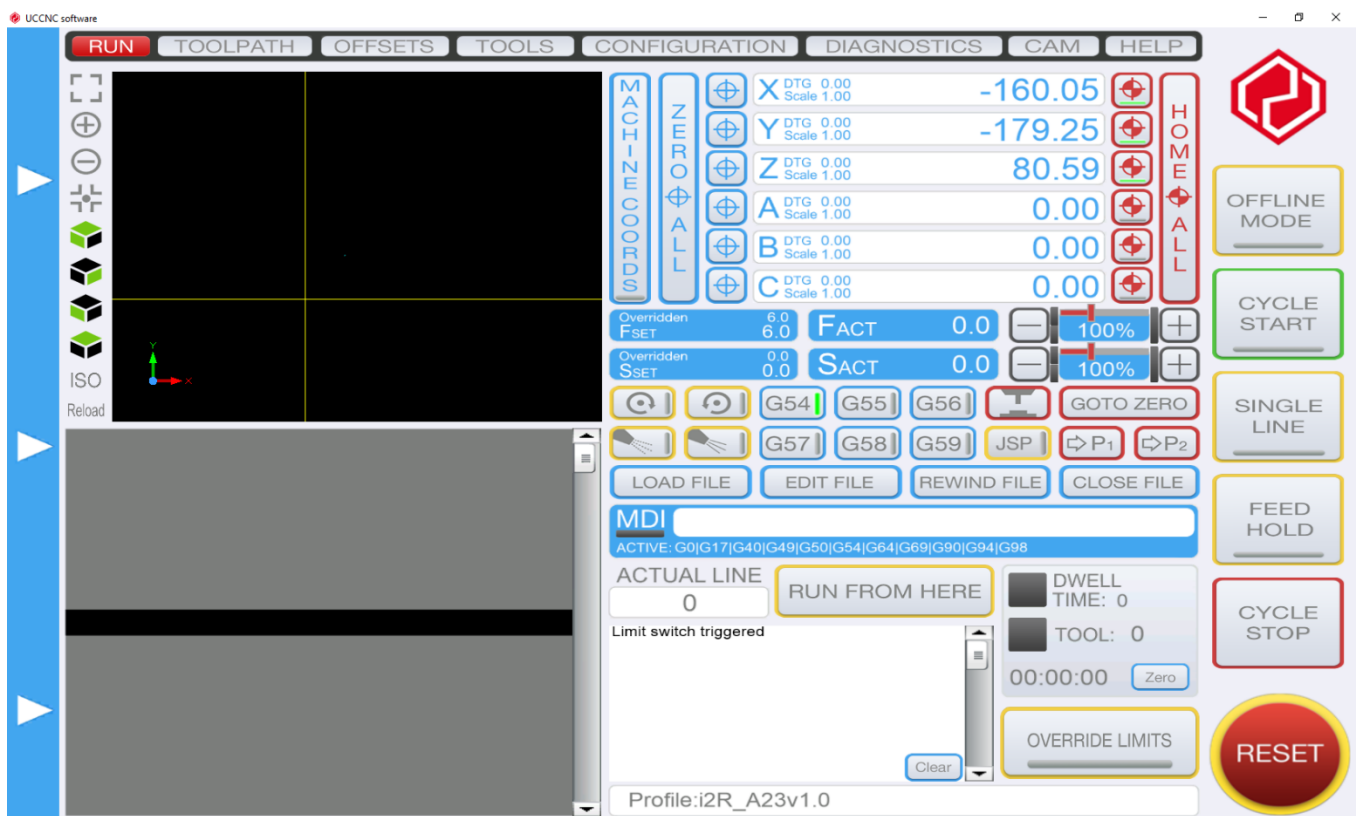
Y extends from the front of the CNC (zero) to the back of the CNC (Y limit)

Z extends from the bed of the CNC to the upper limit of the movement of the head.

The Origin of the Machine Coordinates is located at the front left of the CNC bed.

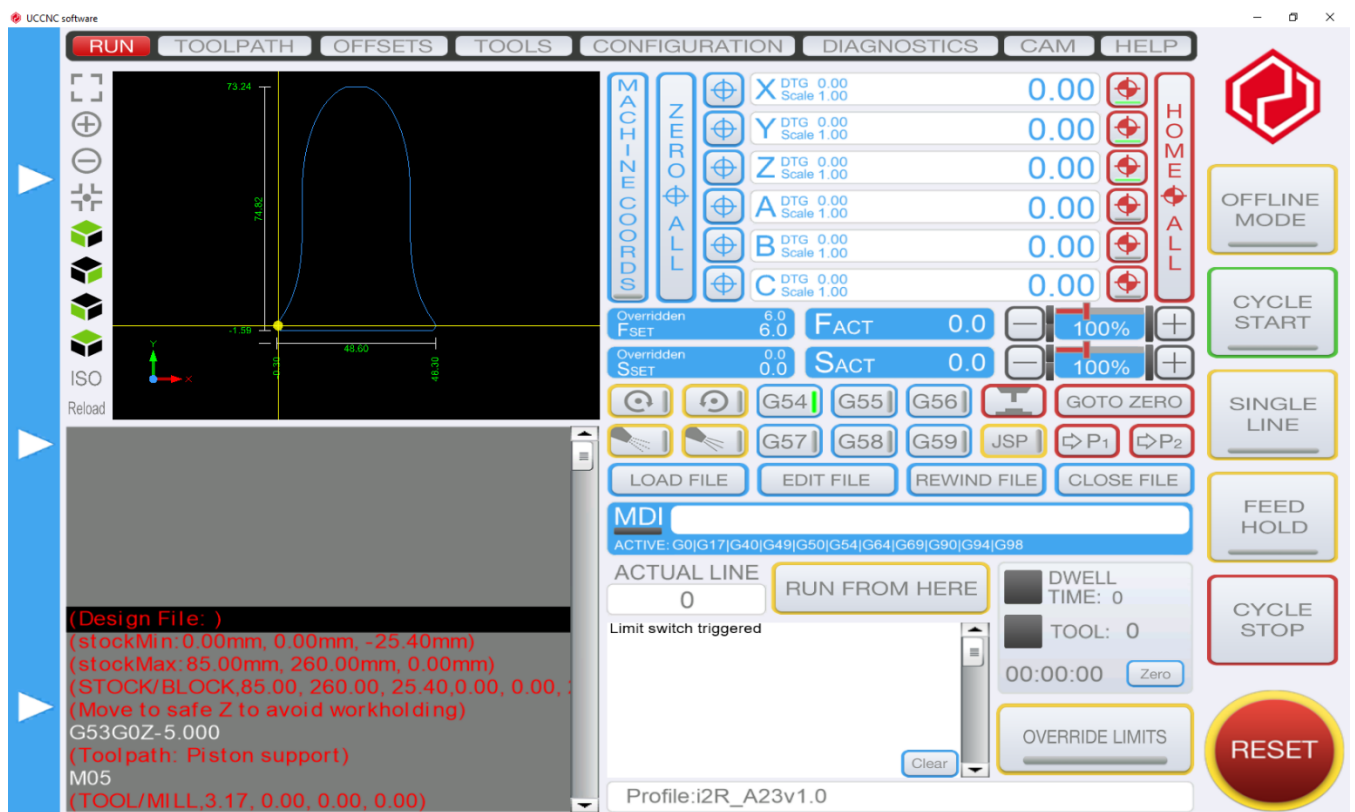
- a. Ensure that the CNC bed is clear of any obstacles.
- b. Click on the **HOME ALL** button on the UCCNC screen. The head will move to the Machine origin at the front left of the bed.
- c. Click on the **ZERO ALL** button to set the Machine Coordinates to zero.

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6. Load your GCode file by clicking on the **Load File** button on the UCCNC screen.
You will see the tool path displayed in the top left of the UCCNC screen, and the GCode displayed in the bottom left of the UCCNC screen.
These 2 displays will dynamically show the progress of the tool head and the steps through the GCode as it is processed. The current XYZ coordinates of the head will be displayed in the top right of the screen.

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7. Clamp the stock material to the bed of the CNC and insert your selected tool in the head. Ensure that the clamps (stays) and stops are correctly placed so that the path of the head will not collide with these clamps. Ensure that all of the clamps (stays) have been screwed down with the same tension to avoid the top of the stock material not being level with the CNC bed.

8. Setting the Work Origin

Jog the head to the starting point of the tool path on the stock material.

Use the keys on the keyboard:

- ← → to move left (down) and right (up) on the X axis
↑ ↓ to move backwards (up) and forwards (down) on the Y axis
PgUp PgDn to move up and down on the Z axis

Each jog key can be pressed momentarily, or sustained for continuous movement. Multiple keys can be pressed for diagonal or combined movement in multiple axes.

9. Moving to the Work Origin.

This can be done, either with a Touch Off pad, or manually with a piece of paper:

- Plug the lead from the Touch Off pad into the socket in the left upright of the gantry, and place the Touch Off pad on top of the stock material and under the tool head. This may require the tool head to be raised in the Z axis (PgUp).

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Click on the **Auto Touch Off** button  on the UCNC screen and the tool head will automatically move down until it touches the Touch Off pad and record this as the Zero origin for the Z axis.

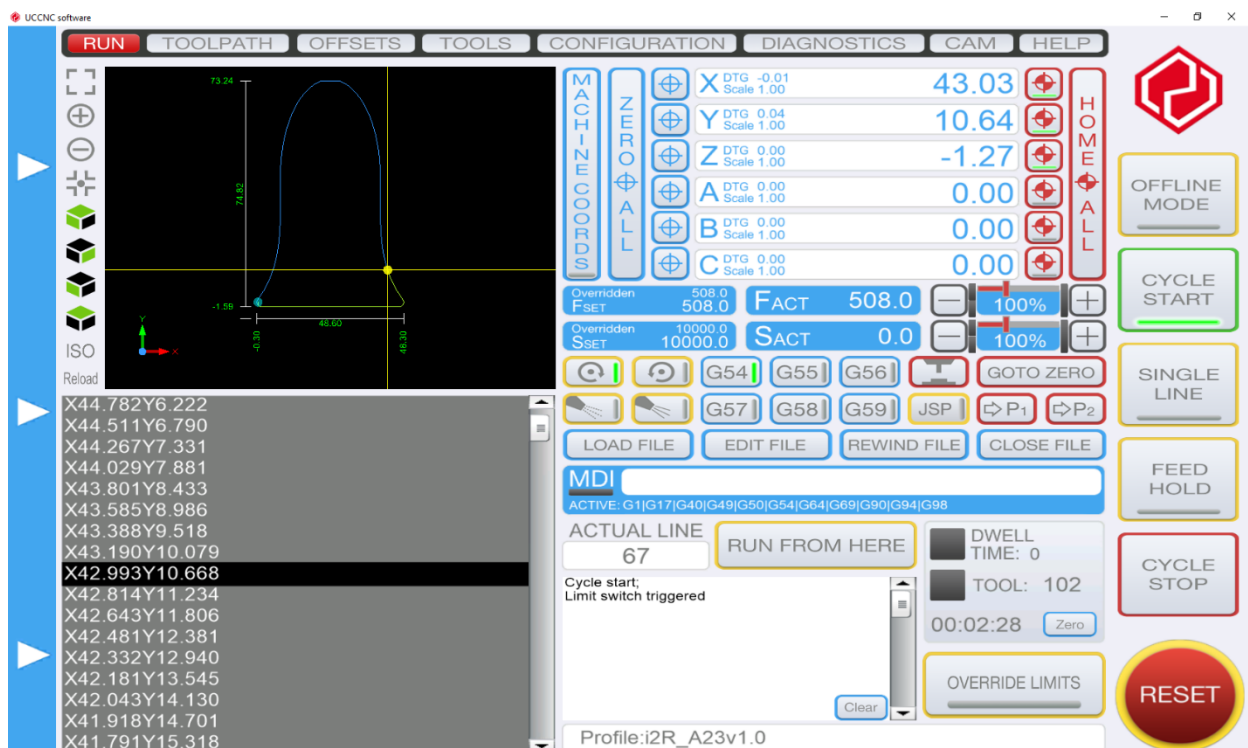
Remove the Touch Off pad.

- b. Place a piece of 80 gsm paper on top of the stock material, and gently lower the tool head until the tool is touching the paper, so that the paper can be moved with some resistance, but is not stuck.

10. Set the origin of the job's Work Coordinates by clicking on the **Zero All** button on the UCNC screen.

11. Start the job in the GCode file by clicking on the **Cycle Start** button. The CNC will commence the cut from the origin of the Work Coordinates, and treat all GCode commands and arguments as being relative to the origin of the Work Coordinates.

As the job progresses you will see the tool head position shown in the top left of the UCCNC screen, and the current Work Coordinates in the top right of the screen.



If anything goes wrong, or is about to go wrong, immediately **HIT the EMERGENCY STOP** button on the CNC controller, or the **RESET** button on the UCCNC screen.

When the job has finished, click on the **Cycle Stop** button to close the job.

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Other useful features of the UCCNC screen:

- a. A job may stop because the CNC software determines that the head will exceed the limits of one or more axes. If this occurs you will need to reposition, or resize the job, and reset the Work Coordinates to avoid hitting these limits.
- b. There is a **Manual Data Input (MDI)** field located in the middle right of the UCNC screen. To manually execute a GCode command, type the one line command into the **MDI** field and hit the Enter key. The CNC will perform the command.
- c. The **SINGLE LINE** button on the menu on the right hand side of the UCCNC screen allows you to step through the GCode, one command at a time – useful for debugging problems in the GCode.
- d. The **FEED HOLD** button allows you to pause the CNC job and then resume. This is not a good idea as the pause may show as an imperfection on your finished work.

More details are contained in the UCNC Manual.

INDUCTION MANUAL: CAPABILITY ASSESSMENT

CAPABILITY ASSESSMENT

MEMBER'S NAME: _____

Knowledge level

5	Capable	☐	Competent to operate the CNC for a non-trivial job
3	Barely Capable	☐	Requires some supervision to operate the CNC for a non-trivial job.
1	Incapable	☐	Restricted to manual tasks only.

Job Planning

5	Capable	☐	Can competently articulate how to plan and execute a job on the CNC.
3	Barely Capable	☐	Can articulate how to plan and execute a job on the CNC, with restrictions.
1	Incapable	☐	Cannot adequately articulate how to plan and execute a job on the CNC.

Physical ability

5	Capable	☐	Can mount stock material, tighten clamps and accurately jog the tool.
3	Barely Capable	☐	Can mount stock material, tighten clamps and jog the tool satisfactorily.
1	Incapable	☐	Not capable of mounting stock material, tightening clamps or jogging the tool.

Safety practice

5	Capable	☐	Completed a non-trivial job on the CNC without any breach of safety guidelines.
3	Barely Capable	☐	Completed a non-trivial job on the CNC with minor breach of safety guidelines.
1	Incapable	☐	Significant breaches of safety guidelines.

Capacity Score _____ out of 20.

Capable	Barely Capable	Incapable
Achieved at least 17 / 20.	Achieved at least Barely Capable in all categories. Some supervision required.	Did not achieve at least Barely Capable in all categories.

ASSESSED BY: _____ DATE: _____

COMMENTS: